

SEMINAR ON ADVANCED QUANTUM MECHANICS

Modern double-slit experiments & interferometers

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Topic 7A: Modern double-slit experiments & interferometers

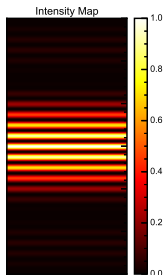
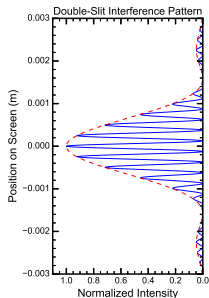
We use the double slit as a guide to single-photon experiments, which-path information, interferometers, molecules, and decoherence.

Central message

**Indistinguishable alternatives
interfere; which-path
information suppresses the
fringes.**

Roadmap

- ① Amplitude addition
- ② Single-photon buildup
- ③ Which-path information
- ④ Interferometers
- ⑤ Molecules and decoherence



Only the essential idea

$$\begin{aligned} P(x) &= |\psi_1(x) + \psi_2(x)|^2 \\ &= |\psi_1|^2 + |\psi_2|^2 + 2\text{Re}[\psi_1^* \psi_2]. \end{aligned}$$

- The cross term is coherence between alternatives.
- It disappears when which-path information exists.

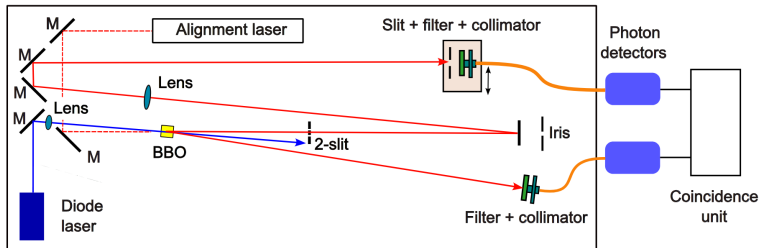
Standard picture

- Fringes follow from phase differences.
- Bright fringes: $d \sin \theta = m\lambda$.
- Spacing: $\Delta x \simeq \lambda L/d$.

Modern emphasis

- How are single photons prepared?
- Which information is actually recorded?
- How is coherence controlled or lost?

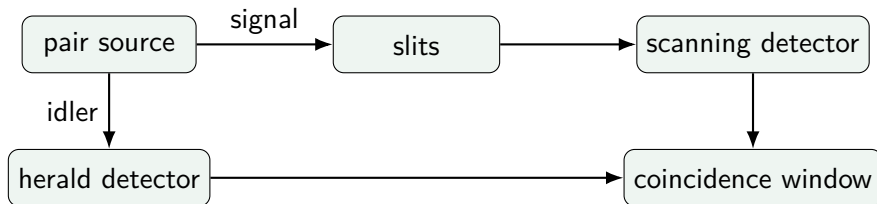
The rest of the talk is about distinguishability and control.



Luo et al.

- Down-conversion creates photon pairs; one photon is the herald.
- The partner photon crosses the slit apparatus and is detected point by point.

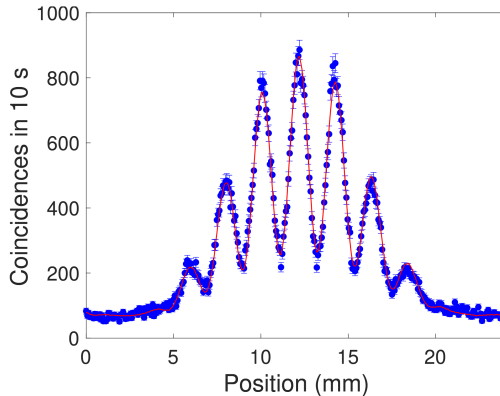
[Luo et al. 2024]



Operational point

A scanning-detector click is accepted only with a herald click in a short time window, suppressing background photons, dark counts, and accidental unpaired events.

[Luo et al. 2024]



Lesson

Each accepted photon is localized; the ensemble reconstructs diffraction and two-path interference statistically.

[Luo et al. 2024]

Without path information,

$$|\psi\rangle = \frac{1}{\sqrt{2}} \left(|1\rangle + e^{i\phi} |2\rangle \right).$$

A marker degree of freedom can store path information:

$$|\Psi\rangle = \frac{1}{\sqrt{2}} \left(|1\rangle \otimes |D_1\rangle + e^{i\phi} |2\rangle \otimes |D_2\rangle \right).$$

Visibility control

Fringe visibility is controlled by $|\langle D_1 | D_2 \rangle|$: identical markers preserve fringes, orthogonal markers suppress them.

Quantities

$$V = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}$$

D quantifies path knowledge.

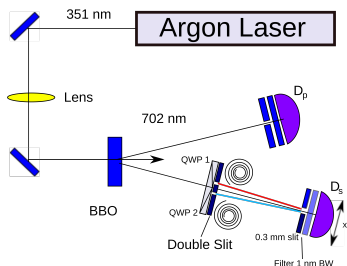
Englert inequality

$$D^2 + V^2 \leq 1.$$

Perfect path knowledge and perfect visibility cannot be obtained in the same data.

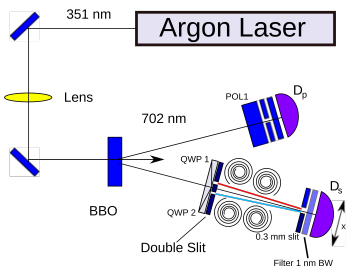
Increasing distinguishability necessarily reduces visible coherence.

[Englert 1996]



Walborn, Cunha, Padua, and Monken, Double slit quantum eraser experiment with polarizer (POL1) removed. In this configuration contrary polarizations prevent interference at D_s .

Path marking: distinguishable alternatives, no unsorted fringes.

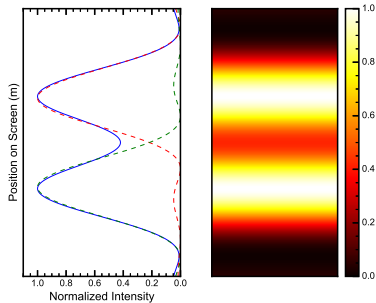


Walborn, Cunha, Padua, and Monken, Double slit quantum eraser experiment with polarizer (POL1) present. In this configuration, interference is detected at D_s .

Erasure: common projection, fringes in selected coincidences.

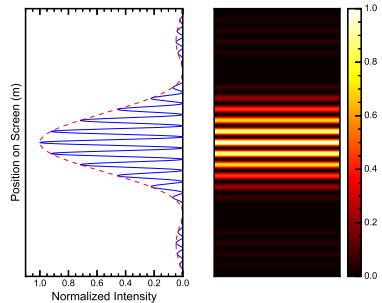
[Luo et al. 2024]

Orthogonal Polarized Double-Slit

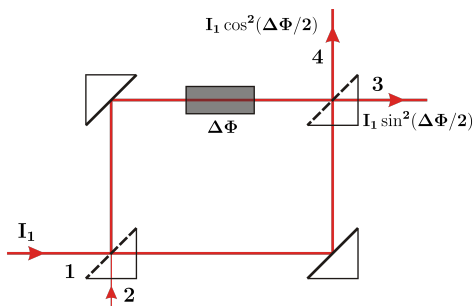


With marker (orthogonal polarization): fringes suppressed.

Generic Double-Slit



No marker (identical polarization): fringes remain.



$$P_1 = \frac{1}{2}(1 + \cos \phi), \quad P_2 = \frac{1}{2}(1 - \cos \phi).$$

Why this is modern

Beam splitters define and recombine the alternatives; phase shifters and detectors let us control coherence and which-path information.

Closed

Second beam splitter present.

- Paths are recombined.
- Output depends on ϕ .
- Interference is visible.

Open

Second beam splitter absent.

- Detectors reveal the path.
- No phase-dependent output.
- Which-path information is available.

Changing the apparatus changes which question the experiment answers.

$$\lambda_{dB} = \frac{h}{p} = \frac{h}{mv}.$$

- Electron interference can also build up one detection at a time.
- C₆₀ fullerene interference extended wave-particle duality to large molecules.
- Later experiments reached molecular masses beyond 25 kDa (kg/mol).

Conceptual point

Large systems do not automatically become classical; they are harder to prepare, isolate, and detect coherently.

[Tonomura et al. 1989; Arndt et al. 1999; Fein et al. 2019]

Small de Broglie wavelength

- larger mass gives smaller λ_{dB}
- narrow velocity distributions are needed
- gratings or optical standing waves must be stable

Many decoherence channels

- collisions with residual gas
- thermal radiation
- vibrations and stray fields

The experimental challenge is preserving coherence long enough to observe fringes.

[Hornberger et al. 2012; Fein et al. 2019]

Let the environment become correlated with the two paths:

$$\frac{1}{\sqrt{2}} (|1\rangle + |2\rangle) \otimes |E_0\rangle \rightarrow \frac{1}{\sqrt{2}} (|1\rangle \otimes |E_1\rangle + |2\rangle \otimes |E_2\rangle).$$

If the environment is not observed, the remaining coherence is proportional to

$$\rho_{12} \propto \langle E_2 | E_1 \rangle.$$

Interpretation

If $|E_1\rangle$ and $|E_2\rangle$ are distinguishable, the environment has effectively measured the path and interference fades.

[Hornberger et al. 2012; Schlosshauer 2007]

Situation	Path information	Interference
Double slit, no detector	unavailable	visible
Path marker at the slits	available	lost/reduced
Eraser with selected data	erased	reappears
Closed Mach-Zehnder	unavailable	phase-dependent
Large molecule with decoherence	stored by environment	lost/reduced

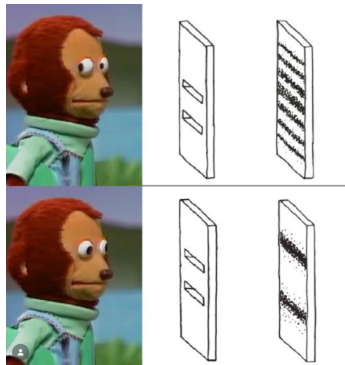
Unifying idea

Interference is suppressed when distinguishability removes coherent amplitude addition.

- Quantum interference is interference of probability amplitudes.
- Single photons, electrons, and molecules show the same buildup logic.
- Which-path information suppresses interference by making alternatives distinguishable.
- Interferometers are controlled phase-sensitive versions of double-slit physics.
- Decoherence explains why interference becomes hard to observe in complex systems.

Indistinguishability creates interference; path information suppresses it.

- [1] B. J. Luo, L. Francis, V. Rodriguez-Fajardo, F. Khoshnoud, and E. J. Galvez, “Young’s double-slit interference demonstration with single photons,” *American Journal of Physics*, 2024.
- [2] A. Tonomura, J. Endo, T. Matsuda, T. Kawasaki, and H. Ezawa, “Demonstration of single-electron buildup of an interference pattern,” *American Journal of Physics* **57**, 117–120 (1989).
- [3] B.-G. Englert, “Fringe visibility and which-way information: An inequality,” *Physical Review Letters* **77**, 2154–2157 (1996).
- [4] M. Arndt et al., “Wave-particle duality of C_{60} molecules,” *Nature* **401**, 680–682 (1999).
- [5] Y. Y. Fein et al., “Quantum superposition of molecules beyond 25 kDa,” *Nature Physics* **15**, 1242–1245 (2019).
- [6] K. Hornberger et al., “Colloquium: Quantum interference of clusters and molecules,” *Reviews of Modern Physics* **84**, 157–173 (2012).



Thank you for your attention!
Any questions?